



3.3 Volume, surface area and geometric applications

Geometry and trigonometry · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Standard formulas give the volume and surface area of prisms, cylinders, cones, pyramids and spheres. Composite solids are handled by adding or subtracting the volumes/surface areas of simpler component shapes. These formulas are often combined with trigonometry (angles of elevation/depression, bearings, slant heights via Pythagoras) to solve applied 3D geometry problems.

Worked example: follow the reasoning

A solid consists of a cylinder of radius 4 cm and height 10 cm with a hemisphere of radius 4 cm removed from the top. Find the volume of the remaining solid, to 1 decimal place.

Solution: $V(\text{cylinder}) = \pi r^2 h = \pi(16)(10) = 160\pi \approx 502.7 \text{ cm}^3$.

$V(\text{hemisphere}) = \frac{2}{3}\pi r^3 = \frac{2}{3}\pi(64) \approx 134.0 \text{ cm}^3$. Remaining volume $\approx 502.7 - 134.0 = 368.6 \text{ cm}^3$.

Quick check · multiple choice

The volume of a cone with radius 3 cm and height 8 cm is:

- A 24π
- B 72π
- C 8π
- D 216π

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A grain silo is modelled as a cylinder of radius 3.5 m and height 8 m, topped with a cone of the same radius and slant height 5 m. (a) Find the height of the cone. (b) Find the total volume of the silo, to the nearest m^3 . (c) Find the total outer surface area of the silo (including the base), to the nearest m^2 . [8]

Final answer: _____



Answers and explanation

Question 1 [8]

(a) height = $\sqrt{5^2 - 3.5^2} = \sqrt{12.75} \approx 3.571$ m. (b) $V(\text{cylinder}) = \pi(3.5^2)(8) = 98\pi \approx 307.88$; $V(\text{cone}) = (1/3)\pi(3.5^2)(3.571) \approx 45.82$; total $\approx 353.7 \approx 354$ m³. (c) Base area = $\pi(3.5^2) \approx 38.48$; cylinder lateral = $2\pi(3.5)(8) = 56\pi \approx 175.93$; cone lateral = $\pi(3.5)(5) = 17.5\pi \approx 54.98$; total ≈ 269 m².

Quick check answer

A. $V = (1/3)\pi r^2 h = (1/3)\pi(9)(8) = 24\pi$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.